

IIT ROPAR AIML COURSE

Machine Learning Performance Metrics

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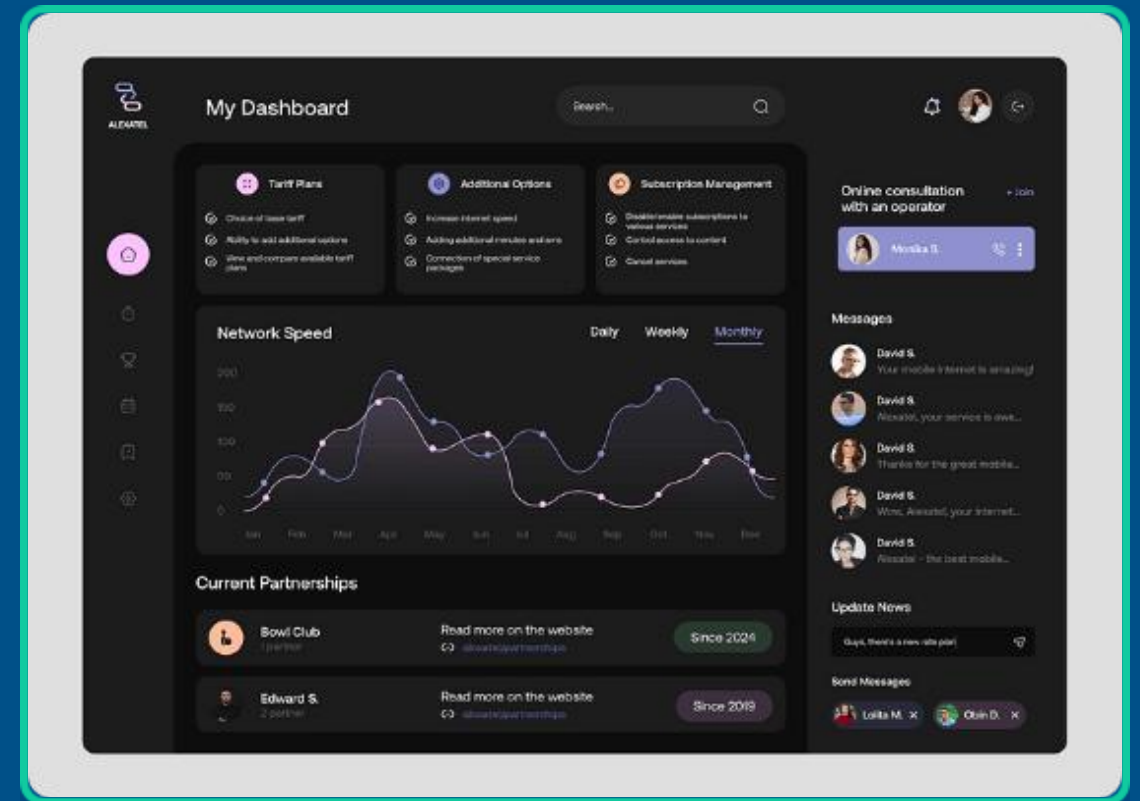


Evaluating Classification Models

Moving Beyond Simple
Accuracy

Why Do We Need Metrics?

- Training a model is only half the battle. We must rigorously **verify its performance** on unseen data.
- Different algorithms (like Logistic Regression) require specific metrics to understand their true predictive power.
- High accuracy can be misleading, especially with imbalanced datasets (e.g., medical diagnoses or fraud detection).
- Metrics help us understand the *types* of errors our model is making.



The Core Evaluation Trio



Confusion Matrix

The foundational $N \times N$ table used to define the performance of a classification algorithm. It reveals the exact counts of correct and incorrect predictions.



Precision

Answers: Out of all the positive predictions made by the model, how many were actually correct? Focuses on minimizing False Positives.



Recall (Sensitivity)

Answers: Out of all the actual positive cases in the data, how many did the model find? Focuses on minimizing False Negatives.

Anatomy of the Confusion Matrix

	Predicted Positive (1)	Predicted Negative (0)
Actual Positive (1)	True Positive (TP) Model correctly predicted the positive class.	False Negative (FN) Type II Error: Model missed a true positive.
Actual Negative (0)	False Positive (FP) Type I Error: Model falsely claimed a positive.	True Negative (TN) Model correctly predicted the negative class.

Calculating Accuracy

When to use it?

- Accuracy represents the ratio of correctly predicted observations to the total observations.
- It is a great measure **only when the dataset is symmetric** (values of false positive and false negatives are almost same).
- Never rely solely on accuracy for imbalanced datasets!

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

The Trade-off: Precision vs. Recall

Precision Formula

$$\text{Precision} = \frac{TP}{TP + FP}$$

Focuses on **False Positives (FP)**. Example: Spam detection. A false positive means a legitimate email goes to the spam folder (highly undesirable).

Recall Formula

$$\text{Recall} = \frac{TP}{TP + FN}$$

Focuses on **False Negatives (FN)**. Example: Cancer detection. A false negative means a sick patient is told they are healthy (fatal consequence).

Advanced Metrics: F-Beta Score

Finding the Perfect
Balance

The F-Beta Score

$$F_{\beta} = (1 + \beta^2) \cdot \frac{\text{Precision} \times \text{Recall}}{(\beta^2 \times \text{Precision}) + \text{Recall}}$$

- **What is it?** The harmonic mean of Precision and Recall.
- **$\beta = 1$ (F1 Score):** False Positives and False Negatives are equally important.
- **$\beta > 1$ (e.g., $\beta = 2$):** Gives more weight to **Recall** (minimizing False Negatives is critical).
- **$\beta < 1$ (e.g., $\beta = 0.5$):** Gives more weight to **Precision** (minimizing False Positives is critical).

Real-World Case: Rain Prediction



Problem: Will it rain tomorrow?

Let's analyze the worst-case scenario for our model's errors:

- **False Positive:** Model predicts rain, but it's sunny. You carry an umbrella for no reason (minor inconvenience).
- **False Negative:** Model predicts sunny, but it rains. You don't carry an umbrella and get soaked (major inconvenience!).

Conclusion: In this scenario, we must focus on minimizing False Negatives. Therefore, we prioritize **Recall** (or use an F-Beta score where $\beta > 1$).

Questions?

Thank you for attending.

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